A CASE-STUDY BASED REPORT

ON

Exploratory Data Analysis of Network Traffic for Intrusion Detection Using the UNSW-NB15 Dataset

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ABSTRACT

This case-study report presents an Exploratory Data Analysis (EDA) of the UNSW-NB15 training dataset for understanding network traffic and identifying patterns associated with normal activity and potential cyber attacks. The dataset used in the project contains 175,341 records and 45 original columns. The analysis examines data quality, feature types, attack labels, attack-category distribution, numerical traffic characteristics, and relationships among selected features. The project preprocessing converts categorical attributes into numerical form, producing 191 encoded features, and standardizes the data before dimensionality reduction. Principal Component Analysis (PCA) reduces the encoded feature space to 30 components, while t-SNE visualizes a sample of 2,000 records in two dimensions. Isolation Forest then flags unusual records for further investigation. The reported pipeline output contains 8,767 potential anomalies out of 175,341 records. These are model-flagged observations, not confirmed attacks.

1. INTRODUCTION

Modern networks generate large volumes of connection records. Manually inspecting every record is difficult, so exploratory analysis and machine-learning techniques can help security teams understand traffic behavior and prioritize unusual observations. EDA is the initial step: it describes the data, checks quality, explores distributions, and identifies relationships that influence later modelling.

This study uses the UNSW-NB15 training dataset in the project “High-Dimensional Network Log Visualization Using PCA and t-SNE.” The EDA findings guide preprocessing and provide context for PCA-based reduction, t-SNE visualization, and Isolation Forest anomaly detection.

2. BACKGROUND

Network-flow datasets represent connections through measurements such as duration, packet counts, byte counts, traffic rates, protocol, and connection state. Some attributes are numeric while others are categorical. Machine-learning methods often require categorical data to be encoded and numeric features to be placed on comparable scales. PCA summarizes variation into fewer components, t-SNE is useful for visualizing local structure in two dimensions, and Isolation Forest detects observations that differ from the majority based on feature patterns.

3. MOTIVATION FOR THE PROJECT

The motivation is to make high-dimensional network logs easier to inspect and to support early discovery of unusual traffic. EDA helps reveal data quality issues, class distribution, and feature relationships before applying dimensionality reduction and anomaly detection. The goal is not to replace security analysts, but to provide an interpretable starting point for further investigation.

4. OVERVIEW OF THE WORK CARRIED OUT

The work includes loading the UNSW-NB15 training data, checking dimensions and data quality, exploring numerical and categorical features, examining labels and attack categories, encoding categorical fields, scaling the features, reducing dimensionality with PCA, visualizing a 2,000-record sample using t-SNE, and applying Isolation Forest to identify potential anomalies. The

generated outputs include two t-SNE plots, an anomaly-results CSV file, and an Isolation Forest anomaly plot.

5. SCOPE AND RESPONSIBLE-USE STATEMENT

This case study is limited to exploratory analysis and unsupervised anomaly detection on the supplied UNSW-NB15 training dataset. Results depend on the dataset, preprocessing decisions, sampling, and model settings. An anomaly score does not prove malicious activity; flagged records require validation using additional evidence and expert review. The analysis should be used for learning and defensive research.

6. PROBLEM STATEMENT

The dataset contains many records and multiple traffic features, making direct inspection and visualization difficult. The problem is to understand the data characteristics and prepare a consistent feature space that can be visualized and examined for unusual patterns without assuming that every flagged record is an attack.

7. CASE-STUDY SCENARIO

A university Security Operations Centre receives network-flow records from its internal network. The team wants an initial view of traffic types and possible unusual connections. Using the UNSW-NB15 training data as an example dataset, the team performs EDA, reduces the feature space, visualizes traffic structure, and uses an unsupervised model to prioritize records for follow-up.

8. CENTRAL CASE-STUDY QUESTION

What useful data-quality, distribution, and feature-relationship insights can be identified in UNSW-NB15, and how can those findings support PCA, t-SNE visualization, and Isolation Forest-based anomaly screening?

9. FORMAL PROBLEM DEFINITION

Given a dataset D containing n network records and p original attributes, characterize the data quality and feature distributions, transform categorical attributes into numeric features, standardize the feature values, and create a lower-dimensional representation for visualization. Finally, use an unsupervised anomaly detector to flag records whose feature patterns differ from the majority. The outputs are descriptive findings, plots, and a list of candidate anomalies for investigation.

10. OBJECTIVES THAT GUIDE THE STUDY

Describe the size, structure, and feature types of the dataset.

Check missing values and duplicate records.

Explore the binary label and attack-category distribution.

Summarize selected numerical network features and their relationships.

Explain why encoding, scaling, and PCA are needed.

Visualize a manageable sample using t-SNE.

Identify potential anomalies using Isolation Forest and state the limitations clearly.

11. ASSUMPTIONS AND LIMITATIONS

The reported dataset and pipeline figures are based on the project execution results supplied for this study. The dataset is treated as a training dataset and may not represent every real network environment. The t-SNE plot uses a 2,000-record sample, so it does not display all 175,341 records. t-SNE distances and clusters are visual aids rather than proof of class separability. The reported 5% Isolation Forest contamination setting influences the number of records flagged; model flags are not ground-truth attack labels.

12. DATASET AND EXPLORATORY DATA ANALYSIS

12.1 About the Dataset

The UNSW-NB15 training dataset used in the project contains 175,341 records and 45 original columns. The data includes numeric traffic measurements and categorical fields such as protocol, service, and state, along with target information including label and attack category. After categorical encoding, the feature matrix expands to 191 columns.

12.2 Purpose of EDA

EDA is performed to understand the dataset before machine learning. It helps verify the integrity of the loaded data, examine class and attack-category distributions, inspect feature ranges, and identify relationships among numerical attributes. These insights inform preprocessing and help prevent misleading interpretation of the later plots and anomaly results.

12.3 Preparation Steps Actually Performed

Loaded the UNSW-NB15 training CSV into a Pandas DataFrame.

Checked dataset dimensions: 175,341 rows and 45 columns.

Checked missing values: 0 reported in the loaded dataset.

Checked duplicate rows: the row count remained 175,341 after duplicate removal.

Encoded categorical attributes, resulting in 191 features.

Applied feature scaling before PCA.

Applied PCA to obtain 30 components.

Selected 2,000 records for t-SNE visualization.

Ran Isolation Forest and saved anomaly predictions and a visualization.

12.4 EDA Exploration – Implementation

The following Pandas commands illustrate the core EDA checks used for the case study. They can be run on the same training CSV file in the project environment.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("UNSW_NB15_training-set.csv")
print("Shape:", df.shape)
print("Missing values:", df.isna().sum().sum())
print("Duplicate rows:", df.duplicated().sum())
print(df.describe(include="all").transpose())

if "label" in df.columns:
    print(df["label"].value_counts())
if "attack_cat" in df.columns:
    print(df["attack_cat"].value_counts(dropna=False))
    df["attack_cat"].value_counts(dropna=False).plot(kind="bar")
    plt.title("Attack Category Distribution")
    plt.tight_layout()
    plt.show()
```

12.5 Summary of EDA Findings

Dataset size: 175,341 records and 45 original columns.

Data quality: 0 missing values and 0 duplicate rows were reported in the project output.

Feature expansion: categorical encoding produced 191 columns.

Scaling: standardization is important because traffic attributes can use different units and ranges.

Dimensionality reduction: PCA reduced the encoded representation from 191 features to 30 components.

Visualization: t-SNE was run on a sample of 2,000 records and produced two saved plots.

Anomaly screening: Isolation Forest flagged 8,767 records as potential anomalies and 166,574 as normal according to the project output.

13. ALGORITHM SELECTION AND IMPLEMENTATION

PCA was selected to reduce the high-dimensional encoded feature space to 30 components. This makes later processing more manageable and provides a compact representation of the data. t-SNE was selected to map the PCA representation of a 2,000-record sample into two dimensions for visual inspection of local patterns. Isolation Forest was selected because it is an

unsupervised method that can flag observations that are easier to isolate from the bulk of the data.

The project output reported 166,574 normal records and 8,767 anomaly records. These values sum to 175,341, and the flagged proportion is approximately 5%. This is a model output affected by the chosen contamination parameter; it should not be interpreted as a verified attack rate.

14. CONCLUSION

The EDA case study provides a structured understanding of the UNSW-NB15 training dataset and its readiness for machine-learning analysis. It verifies the reported dataset dimensions and data quality, examines mixed feature types and attack labels, and explains the need for encoding, scaling, and dimensionality reduction. The project then uses PCA and t-SNE for compact visualization and Isolation Forest for unsupervised anomaly screening. The resulting anomalies are candidates for further analysis rather than confirmed attacks. Future work can compare predictions with known labels, evaluate model performance using suitable metrics, and test the workflow on additional network traffic.

REFERENCES

Moustafa, N. and Slay, J. (2015). UNSW-NB15: a comprehensive data set for network intrusion detection systems.

Scikit-learn documentation: Principal Component Analysis (PCA), t-SNE, StandardScaler, and Isolation Forest.

Pandas documentation: DataFrame inspection, descriptive statistics, and missing-value analysis.